Proposal: Characterizing Deception in Large Language Models via Latent Spectral Dynamics

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Abstract

As the capabilities of Large Language Models (LLMs) scale, alignment failures such as strategic deception and sycophancy present critical safety risks. Current mechanistic interpretability approaches largely analyze these models statically, missing the temporal evolution of hidden states during autoregressive generation. In this work, we propose extending the *Latent Spectral Dynamics* framework to characterize deception. Building upon its recent application in analyzing hallucinations, we model the token-wise residual stream as a stochastic, autonomous, discrete-time dynamical system. We lift these nonlinear dynamics into an infinite-dimensional linear space using Koopman operator theory. To overcome the computational bottleneck of our current matrix-factorization-based Hankel-EDMD approach, we propose a deep learning paradigm, training Deep Koopman Autoencoders to scale spectral analysis to larger models. Ultimately, we hypothesize that the Markovian transition in the dynamics will persist and strategically deceptive trajectories will exhibit distinct spectral signatures, distinguishing them from both factual and uncertain generation.

1 Introduction and Motivation

Large Language Models (LLMs) are increasingly deployed as autonomous agents and reasoning engines. However, this increased capability brings sophisticated alignment failures. While *hallucination* (generating fluent but factually ungrounded text) is well-documented, a more insidious failure mode is *deception*. Deceptive models may strategically output false information to achieve a goal (Scheurer et al., 2023), hide their true capabilities to evade safety protocols (Hubinger et al., 2024), or sycophantically align with user biases despite retaining internal knowledge of the truth.

The primary motivation for this project is to characterize these emergent deceptive behaviors through the lens of computational dynamical systems. This project extends upon our current work in Mechanistic Interpretability (currently under review for COLM 2026), where we successfully formulated LLM autoregressive generation as an autonomous discrete-time dynamical system. In that work, we utilized Koopman operator theory and Hankel-Extended Dynamic Mode Decomposition (Hankel-EDMD) to demonstrate that factual and hallucinated regimes possess distinctly identifiable spectral signatures on the approximated Koopman operators.

However, Hankel-EDMD relies on exact matrix factorizations over massive, uncompressed coordinate bases, making it prohibitively expensive in CPU and RAM when scaling to larger models (e.g., 32B+ parameters). Furthermore, deception is a fundamentally more complex phenomenon than hallucination; it requires the model to maintain a dual-state representation (the hidden truth vs. the deceptive output).

Our objectives for this course project are thus threefold:

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1. *Algorithmic Extension:* We aim to bypass the EDMD matrix factorization bottleneck by implementing a Deep Koopman Autoencoder to learn the spectral dynamics efficiently.
2. *Phenomenological Extension:* We will apply this upgraded framework to datasets designed to invoke strategic deception, analyzing the global latent dynamics to identify the unique spectral motif of deceptive alignment.
3. *Narrative Shift:* We will develop the theoretical language to interpret deception formally from a dynamics angle—framing it as a geometrically distinct attractor or invariant subspace within the activation manifold.

The proposer promises heavy individual effort throughout this extension project.

2 Related Work

This project bridges mechanistic interpretability, dynamical systems theory, and AI safety:

1. **Strategic Deception in LLMs:** Scheurer et al. (2023) demonstrated that LLMs can strategically deceive users (e.g., insider trading scenarios) when put under pressure, providing the behavioral framing for our work.
2. **Sleeper Agents:** Hubinger et al. (2024) highlighted the persistent danger of deceptive alignment, showing that models can harbor hidden deceptive behaviors that evade standard safety fine-tuning.
3. **Universal Motifs in LLM Deception:** Yang and Buzsaki (2025) recently characterized deception across multiple LLM families, identifying a universal motif consisting of three iterative refinement stages in the latent representation. Their success in controlling outputs by steering these specific stages strongly supports our hypothesis that deception manifests as a distinctly identifiable temporal and geometric structure within the activation manifold.
4. **Measure-Theoretic Phase Space Reconstruction:** Botvinick-Greenhouse et al. (2025) proved that, for compact sub-manifolds observed through a generic Lipschitz map, time-delay embeddings preserve the invariant measure of the underlying dynamics. This theorem provides the formal guarantee that our delay-embedded residual stream states $\mathbf{H}_k$ faithfully represent the high-dimensional KV-cache Markov chain, enabling spectral analysis via the Koopman operator.
5. **Deep Koopman Autoencoders:** Lusch et al. (2018) introduced a deep learning framework for discovering universal linear embeddings of nonlinear dynamics, which we adapt to replace our algebraic Hankel-EDMD solver.

3 Methodology

Intuition. At a high level, our approach treats each token-generation step of an LLM as a single timestep in a dynamical system. Just as a physicist might study the trajectory of a particle to infer the forces acting on it, we study the trajectory of an LLM's internal hidden states to infer the generative "regime" it is operating in. We then apply standard tools from dynamical systems theory—specifically, spectral decomposition—to look for distinct patterns (signatures) that separate deceptive generation from honest generation. The formal machinery below makes this intuition precise.

3.1 Markovian Dynamics and Phase Space Reconstruction

Standard analyses treat autoregressive generation as a highly non-Markovian process, factorizing the joint probability as $P_\phi(x_{1:N}) = \prod_{k=1}^{N} P_\phi(x_k \mid x_{<k})$. However, the Transformer architecture provides an exact Markov state through its internal Key-Value (KV) cache. At generation step k, the cache S_k contains all key-value pairs across layers and positions; a deterministic forward pass produces the residual stream $h_k = f_\phi^{(L)}(S_k) \in \mathbb{R}^d$, from which the next token is sampled. Since S_{k+1} is a deterministic function of (S_k, x_{k+1}) and x_{k+1} depends only on S_k, the sequence $\{S_k\}$ is an exact Markov chain:

$$P_\phi(x_{k+1} \mid x_{1:k}) = P_\phi(x_{k+1} \mid S_k) \tag{1}$$

Although S_k is an exact Markov state, it is impractically high-dimensional for operator estimation. We construct a fixed-dimensional surrogate via *Hankel time-delay embedding* of the final-layer residual stream $h_k \in \mathbb{R}^d$, forming $\mathbf{H}_k = [h_{k-W+1}^\top, \ldots, h_k^\top]^\top \in \mathbb{R}^{Wd}$. Under a fixed prompting regime θ^*, generation trajectories concentrate on a low-dimensional sub-manifold of intrinsic dimension $m_{\text{eff}} \ll D_{\text{max}}$. The measure-theoretic phase-space reconstruction theorem (Botvinick-Greenhouse et al., 2025) guarantees that, for generic observation maps and sufficient window size W, the delay-embedded state $\mathbf{H}_k$ preserves the statistical structure of S_k, so that $P_\phi(x_{k+1} \mid S_k) \approx P_\phi(x_{k+1} \mid \mathbf{H}_k)$. In our prior work, we validated this approximation empirically: coordinate-basis Koopman operators achieved stable autonomous rollouts through horizon $h = 8$, confirming that $\mathbf{H}_k$ captures the dominant Markovian dynamics.

3.2 Autonomous Dynamics Formulation

To obtain a self-contained dynamical system on a continuous state space, we must eliminate the intermediate discrete token x_k. Under the approximate state representation above, the next-token probability is the temperature-scaled softmax of the logit vector $l(h_k) = W_U h_k + b_U \in \mathbb{R}^{|\mathcal{V}|}$. We apply the *Gumbel-Max reparameterization*: letting $\mathbf{\Xi}_k \sim \text{Gumbel}(0, \mathbf{I}_{|\mathcal{V}|})$ be exogenous noise, the sampled token becomes a deterministic function $x_{k+1} = \arg\max_{v \in \mathcal{V}}[l(h_k)_v/\tau + \Xi_{k,v}] \equiv \mathcal{T}_{\tau,\phi}(\mathbf{H}_k, \mathbf{\Xi}_k)$. Substituting this into the Transformer's deterministic forward pass and shifting the Hankel window eliminates the discrete token entirely, yielding a **stochastic autonomous discrete-time dynamical system**:

$$\mathbf{H}_{k+1} = \mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k, \mathbf{\Xi}_k) \tag{2}$$

where θ^* indexes the prompting regime. The sequence $\{\mathbf{H}_k\}$ forms an approximate Markov chain on $\mathbb{R}^{Wd}$, governed by the transition kernel $\mathcal{P}_{\theta^*}(d\mathbf{H}_{k+1} \mid \mathbf{H}_k) = \mathbb{P}_{\mathbf{\Xi}}(\mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k, \mathbf{\Xi}_k) \in d\mathbf{H}_{k+1})$. In the greedy limit $\tau \to 0^+$, the stochastic forcing collapses and the system becomes purely deterministic.

3.3 The Koopman Operator and Deep Learning Extension

We lift the dynamics into an infinite-dimensional Hilbert space via the Koopman operator $\mathcal{K}_{\theta^*}$, defined by conditional expectation over the Markov kernel: $(\mathcal{K}_{\theta^*} g)(\mathbf{H}_k) = \mathbb{E}_{\mathbf{H}_{k+1} \sim \mathcal{P}_{\theta^*}(\cdot \mid \mathbf{H}_k)}[g(\mathbf{H}_{k+1})]$.

While our prior work approximated $\mathcal{K}_{\theta^*}$ via Hankel-EDMD, the Gram matrix construction limits scaling. Thus, we propose replacing EDMD with a **Deep Koopman Autoencoder** (Lusch et al., 2018). We define an encoder $\psi : \mathbb{R}^{Wd} \to \mathbb{R}^m$, decoder $\phi : \mathbb{R}^m \to \mathbb{R}^{Wd}$, and a learned Koopman matrix $\mathbf{K} \in \mathbb{R}^{m \times m}$, optimized via:

$$\mathcal{L} = \underbrace{\|\mathbf{H}_k - \phi(\psi(\mathbf{H}_k))\|^2}_{\text{Reconstruction}} + \alpha \underbrace{\|\psi(\mathbf{H}_{k+1}) - \mathbf{K}\psi(\mathbf{H}_k)\|^2}_{\text{Linear Dynamics}} + \beta \|\mathbf{H}_{k+1} - \phi(\mathbf{K}\psi(\mathbf{H}_k))\|^2 \tag{3}$$

This parametric approach decouples sequence length from memory constraints, enabling scalable extraction of deceptive spectral properties (eigenvalues and invariant subspaces).

4 Feasibility and Experimental Design

This proposal builds on an existing codebase co-authored by the proposer; the HuggingFace extraction pipeline and delay-embedding infrastructure are already validated.

Phase 1: Autoencoder Validation (Go/No-Go). We train the autoencoder on the same hallucination trajectories from our COLM submission and verify: (1) multi-step rollout MSE comparable to EDMD, and (2) recovery of the same spectral separation between factual and hallucinated regimes. Only upon achieving spectral parity do we proceed, ensuring any negative result on deception reflects the phenomenon, not methodological failure.

Phase 2: Deception Analysis. We adapt the Apollo Research "Insider Trading" dataset (Scheurer et al., 2023) and additionally evaluate on a Sycophancy dataset to test cross-mode generalization. We extract final-layer residual stream trajectories (W=5) from Qwen2.5-14B/32B/72B across a temperature gradient. The autoencoder is optimized with AdamW; we ablate α to ensure the network learns temporal dynamics rather than memorizing static representations.

Evaluation Metrics & Baselines: We will evaluate the framework via:

1. *Predictability:* We will measure multi-step rollout Mean Squared Error (MSE) on held-out trajectories (i.e., $\mathbf{H}_{k+n} \approx \phi(\mathbf{K}^n \psi(\mathbf{H}_k))$) to ensure the Koopman matrix $\mathbf{K}$ accurately captures the autonomous n-step Markovian dynamics.
2. *Spectral Signatures:* Eigendecomposition of $\mathbf{K}$ to extract eigenvalues λ and spatial modes. Orthogonal eigenmodes will verify that deception occupies a distinctly separable invariant subspace within the activation manifold.
3. *Static Probing Baselines:* We will benchmark our spectral anomaly detection against standard static probing (e.g., training a linear logistic regression classifier on isolated snapshots h_k). We hypothesize that tracking the dynamic Koopman eigenvalues will identify deceptive trajectories earlier and more robustly out-of-distribution compared to static classification.

5 Project Planning

Our timeline is structured in two explicit phases to isolate algorithmic validation from phenomenological extension:

- **Week 2–3:** *Phase 1 — Autoencoder Validation.* Implement the Deep Koopman Autoencoder and train it on our existing hallucination trajectories (Qwen2.5-14B). Validate spectral parity with the prior Hankel-EDMD results: confirm that the autoencoder recovers the same eigenvalue separation between factual and hallucinated regimes and achieves comparable multi-step rollout MSE. This serves as the formal go/no-go gate.
- **Week 4:** *Deception Dataset & Extraction.* Adapt the insider trading dataset and extract final-layer residual stream trajectories for truthful vs. deceptive responses (Qwen2.5-14B). Train static probing baselines.
- **Week 5–6:** *Phase 2 — Scale & Spectral Comparison.* Run the validated autoencoder pipeline on the deception trajectories. Scale to larger models (Qwen2.5-32B/72B). Extract eigenvalues λ and spatial modes from the learned matrix $\mathbf{K}$. Quantify spectral differences between truthful and deceptive regimes. Test generalization on sycophancy data.
- **Week 7-9:** *Drafting.* Consolidate findings, generate phase-plane visualizations, and finalize paper.

LLM Usage

LLMs (Gemini and Claude) were utilized during the drafting of this proposal for paragraph revision, academic tone polishing, and structuring the LaTeX document formatting.

References

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